

Ritik Tiwary

B.Tech – Information and Communication Technology with minor in Computational Science

✉ [Gmail](#)  [Linkedin](#)  [Github](#)

Education

Dhirubhai Ambani University <i>CPI: 7.98</i>	2022 - 2026 Gandhinagar, Gujarat
12th Kendriya Vidyalaya (CBSE) <i>Percentage: 96.40 %</i>	2021- 2022 Gandhinagar, Gujarat
10th Kendriya Vidyalaya (CBSE) <i>Percentage: 94.00%</i>	2019 - 2020 Gandhinagar, Gujarat

Experience

Scale AI (Freelance) <i>AI Model Trainer – Machine Learning Engineer</i>	Jan 2025 – Apr 2025 Remote
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- Collaborated with cross-functional teams to enhance **generative AI models** and **large language models (LLMs)** by solving complex programming problems and debugging code in **Java, Python, and C++**.
- Increased LLM accuracy by crafting and correcting **chain-of-thought reasoning** for algorithmic optimization and logical problem-solving challenges.
- Provided **high-quality feedback, data annotation**, and ground-truth validation to improve model accuracy, safety, and performance across multiple programming languages and frameworks.

Projects

Freelancing Platform Database System <i>PostgreSQL, PgAdmin, SQL, Database Design, C++, DBMS</i>	
• Designed and implemented a comprehensive relational database system for a freelancing platform with normalized schema to BCNF using custom C++ decomposition algorithms and database optimization techniques. • Developed production-ready database solutions using PostgreSQL, performed DDL/DML operations, and created optimized SQL queries with indexing strategies for scalable data operations and performance tuning. • Applied database security measures, query optimization, and performance monitoring for enterprise-grade data management solutions.	
Insight Copilot – Natural Language to SQL Assistant <i>Spring Boot, LangGraph, MongoDB, REST APIs</i>	
• Developed a backend system using Spring Boot that converts natural language input into executable SQL queries and auto-generates visualizations (bar, line, pie). • Integrated LangGraph and RAG (Retrieval-Augmented Generation) for context-aware query generation based on dynamic schema retrieval. • Designed and enforced secure RESTful APIs with JWT-based authentication, role-based access control, and CRUD operations. • Utilized MongoDB for managing dynamic schema structures and user-level metadata in a NoSQL format.	

Technical Skills

Languages: C, C++, Python, Java(basic), SQL
Developer Tools: Git, GitHub, VS Code, PgAdmin, Postman, Figma, Vertex AI Agent
Frameworks & Libraries: Spring Boot, LangChain, LangGraph
Databases & Cloud: PostgreSQL, MySQL, MongoDB, Google Cloud (Vertex AI)
Software Engineering: Agile Development, Version Control, API Testing, Debugging, Documentation
Coursework: Data Structures and Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks, Distributed Systems, System Design, Modeling and Simulation, High-Performance Computing (HPC)
Interests: Generative AI (RAG, Agents), Backend Development, System Design, Competitive Programming

Position Of Responsibility

Associate Core Member

April 2023 – December 2024

Debating Society, DAU

- * Achieved **Top 10 ranking** in Regionals at the **THE IIT Bombay Debate 2024**, showcasing critical thinking and public speaking skills.
- * Led the organization and execution of multiple **intra-college debating events**, and contributed as an **Adjudicator** to evaluate participant performance.

Achievements & Certifications

- **Global Rank 115** in CodeChef Starters Division 2 (195) - Competitive Programming Excellence.
- **Specialist Rating** on Codeforces and **Knight** on LeetCode - Advanced Problem Solving Skills.
- Selected among top **5% applicants** for **Amazon Hackout 2025** based on exceptional coding round performance and technical assessment.
- **Top 1% percentile** in Joint Entrance Examination – Main (JEE-Main) and **Top 1.5%** in Central Board of Secondary Education (CBSE) - Academic Excellence.
- Demonstrated expertise in **algorithmic problem solving**, **data structures**, and **competitive programming** with consistent high performance.